

Simulation Input Identification Module (SIIM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Type: Parent Standard Module

Governed Space: Simulation Input Identification

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AI-Readable: Yes

Authority: OOF®

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Canonical Language: English (UCL™)

Governed Space

Simulation Input Identification

Canonical Definition

Simulation Input Identification Module (SIIM) defines the governance framework through which every materially relevant data object, value, signal, measurement, state, distribution, event, record, synthetic input, derived input, human-supplied input, model-generated input, or other information capable of influencing a simulation is identified, classified, distinguished, source-referenced, bound to its simulation role, and maintained as a traceable Simulation Input.

SIIM establishes the first internal governance space of SIDS by determining what is entering the simulation before questions of representativeness, transformation, temporal applicability, and uncertainty are evaluated.

Operational Role

SIIM serves as the first internal governance space of the Simulation Input & Data Standard (SIDS).

The preceding Parent Standard, Simulation Model & Environment Standard (SMES), establishes the controlled model and environment through which simulation

occurs.

SIIM asks:

What information is proposed to enter that governed simulation structure, what exactly is each input, where does it originate, and what role can it influence?

The governed progression is:

Governed Simulation Model & Environment → Input Requirement → Proposed Input → Input Identification → Input Classification → Origin Reference → Simulation Role Binding → Simulation Input Register

The resulting identified input basis becomes available to the subsequent SIDS modules for representativeness, transformation, temporal, and uncertainty governance.

Module Operational Space

SIIM governs: simulation input discovery, input identification, input naming, input distinction, input classification, input origin reference, source association, input type, input format, input unit, input semantics, input state, input role, input ownership reference where relevant, observed inputs, measured inputs, historical inputs, live inputs, derived inputs, estimated inputs, synthetic inputs, human-supplied inputs, externally supplied inputs, model-generated inputs, scenario-specified inputs, input-to-object binding, input-to-model binding, input-to-environment binding, input-to-configuration binding, input-to-scenario reference, input dependencies, composite inputs, duplicate inputs, ambiguous inputs, unidentified inputs, input substitution visibility, input-set membership, input identifiers, and input traceability.

Module Function

SIIM exists to prevent a simulation from consuming materially influential information whose identity or role is unclear.

It prevents: unidentified inputs entering simulation execution, ambiguous variables being treated as equivalent, different data objects sharing an indistinguishable identity, source origin disappearing after ingestion, synthetic information appearing as observed information, derived information appearing as directly measured information, historical information appearing as live information, units or semantics becoming detached from values, input roles becoming unclear, substitutions occurring invisibly, input dependencies being hidden, and simulation outputs becoming impossible to reconstruct because the actual consumed inputs cannot be determined.

Simulation Input

A Simulation Input is any identifiable information object capable of materially influencing: simulation initialization, model behavior, environmental state, actor behavior, state transitions, simulation progression, scenario conditions, or simulation outputs.

A Simulation Input may be:

Data

Value

Signal

Measurement

State

Event

Distribution

Record

Sequence

Instruction

External Observation

Synthetic Information

or another governed information object consumed by the simulation.

Material Input

SIIM focuses on material Simulation Inputs.

An input is material where changing, removing, substituting, or misidentifying it could materially affect: simulation configuration, simulation behavior, scenario conditions, execution, output, interpretation, or downstream reliance.

Not every technical field requires equal governance depth.

Input Requirement

Before an input is identified, the simulation may establish an Input Requirement.

An Input Requirement defines what information the simulation needs without presupposing which specific data object will satisfy that need.

For example:

Requirement: Current external temperature

may later be satisfied by:

Sensor TEMP-04 measurement

This distinction prevents the required information from being confused with the particular source selected to provide it.

Input Identification

Every material Simulation Input should be sufficiently identifiable before governed simulation use.

Identification may establish: Simulation Input Identifier, input name, input description, input type, input origin, source reference, unit, format, semantic meaning, temporal reference where known, transformation state, applicable Simulation Object, applicable model or environment element, and intended simulation role.

Simulation Input Identifier

A Simulation Input Identifier provides a persistent reference through which a material input can remain distinguishable throughout the simulation lifecycle.

Conceptually:

SI-0001

may identify one governed input object.

The identifier should remain connected to the actual input rather than merely to a display label. Identity $\neq$ Variable Name

SIMULOS® establishes:

A variable name is not sufficient input identity where materially different information can exist under that same name.

For example:

Temperature

could refer to: external temperature, engine temperature, battery temperature, historical average temperature, predicted temperature, or synthetic scenario temperature.

SIIM requires sufficient distinction.

Input Naming

Input names should communicate the represented information with enough specificity to prevent material ambiguity.

Where necessary:

Temperature

should become:

External Ambient Temperature

or another sufficiently precise representation.

Naming alone does not establish full identity, but poor naming can weaken traceability.

Input Semantics

SIIM governs what the input means, not only how it is technically encoded.

For example:

0.75

may represent: 75% probability, normalized sensor output, friction coefficient, model confidence, actuator position, or another quantity.

The numerical value alone is insufficient.

Input Units

Where units materially affect interpretation, they must remain associated with the input.

For example:

50

is not a sufficiently defined physical input if it could mean:

50 km/h

50 mph

50 °C

or:

50 milliseconds.

Input Format

Input format may include: scalar, vector, matrix, categorical value, time series, event stream, image, audio, text, geospatial data, structured record, distribution, or another machine- or human-readable form.

Format should remain identifiable where it affects simulation interpretation.

Input Classification

SIIM classifies Simulation Inputs according to their material origin and role.

Possible classes include: Observed Input

Information obtained through observation of represented reality.

Measured Input

Information produced through a measurement process.

Historical Input

Information originating from a previous period.

Live Input

Information supplied from an active operational environment during or near simulation execution.

Estimated Input

Information established through estimation where direct measurement or observation is unavailable.

Derived Input

Information calculated from other information objects.

Synthetic Input

Information generated rather than directly observed from the represented reality.

Human-Supplied Input

Information introduced by a human participant or operator.

Externally Supplied Input

Information supplied by an external organization, system, service, or data provider.

Model-Generated Input

Information produced by another model and consumed by the simulation.

Scenario-Specified Input

Information deliberately introduced to establish a scenario condition.

One input may require multiple classifications where its provenance or role is composite.

Classification Must Follow Reality

SIMULOS® establishes:

Input classification must describe how the information was actually produced, not how the simulation wishes to interpret it.

A model-generated estimate does not become a measured input merely because it

is formatted like sensor data.

Input Origin

Input Origin identifies the point from which the information entered the governed simulation chain.

Origin may refer to: sensor, database, organization, person, operational system, AI system, external model, simulator, scientific instrument, historical dataset, synthetic generator, or another identifiable origin.

Origin ≠ Reliability

SIIM identifies origin.

It does not establish that the origin is reliable.

Source reliability may require governance under VALIDOS® or another applicable architecture.

The distinction is important:

Known Origin ≠ Trusted Origin Preservation

Where materially relevant, origin should remain attached throughout:

Origin → Input → Transformation → Simulation Use

Subsequent transformation should not silently erase the origin of the information.

Source Reference

An input may maintain a reference to the source object or system from which it originates.

The source reference may include: source identifier, system identifier, dataset identifier, sensor identifier, provider identifier, model identifier, or another persistent reference.

Input State

An input may exist in different states.

For example:

Raw

Cleaned

Normalized

Aggregated

Derived

Imputed

Synthetic

or:

Transformed

SIIM identifies the state.

SDTM — Simulation Data Transformation Module governs the transformation itself.

Raw Input

A Raw Simulation Input is information before simulation-specific transformation.

Raw does not mean: correct, reliable, representative, complete, or suitable.

It only describes its position relative to subsequent processing.

Derived Input

A Derived Input should preserve the relationship:

Source Input(s) → Derivation → Derived Input

SIIM identifies the resulting object as derived.

SDTM governs the methodological transformation.

Synthetic Input Identification

Synthetic inputs must remain distinguishable from directly observed or measured information.

SIMULOS® establishes:

Synthetic information must never acquire an observed-data identity merely because it is technically indistinguishable in format.

Estimated Input Identification

Estimated values should remain identifiable as estimates.

For example:

Estimated Vehicle Mass = 1,850 kg

should not silently become:

Measured Vehicle Mass = 1,850 kg

after ingestion.

Live vs Historical Input

SIIM distinguishes whether information originates from:

current or near-current operational conditions

or:

historical records.

Detailed temporal suitability is governed by STDM — Simulation Temporal Data Module.

SIIM preserves the initial classification required for that assessment.

Input Role

An input should remain connected to the role it performs in the simulation.

Possible roles include: initialization, state update, environmental condition, actor condition, control signal, disturbance, behavioral input, scenario condition, external event, calibration reference, or another defined simulation function.

Input-to-Simulation Object Binding

Where relevant:

Simulation Input → Simulation Object

should remain identifiable.

This prevents inputs associated with one Simulation Object from being silently applied to another.

Input-to-Model Binding

Where an input affects a specific model component:

Simulation Input → Model Component

should remain traceable.

This supports later analysis of which information could have influenced which simulated behavior.

Input-to-Environment Binding

Inputs affecting environmental conditions should preserve:

Simulation Input → Environment Element / State

For example:

Rainfall Input → Road Surface Environment State Input-to-Configuration Binding

Where an input is valid only under a particular simulation configuration, that relationship should remain visible.

Input-to-Scenario Reference

An input may be associated with one or more scenarios.

SIIM records the association where known.

SSS — Simulation Scenario Standard governs scenario design and coverage.

Input Dependency

One input may depend upon another.

For example:

Vehicle Speed

may contribute to:

Calculated Braking Distance Input

The dependency should remain visible where it materially affects traceability.

Composite Input

A Composite Input combines multiple information objects into one simulation-consumed input.

Conceptually:

Input A + Input B + Input C → Composite Input D

The composite should not erase the identities of its materially relevant components.

Duplicate Input

Two technically separate inputs may represent the same underlying information.

Duplicate inputs can create: double counting, unintended weighting, correlated

influence, or false independence.

SIIM identifies material duplication for subsequent treatment.

Ambiguous Input

An Ambiguous Simulation Input exists where the available information is insufficient to determine exactly: what the input represents, where it originated, which unit applies, which object it describes, or which role it performs.

Material ambiguity should be resolved or explicitly preserved before simulation use.

Unidentified Input

An Unidentified Simulation Input is a materially influential input whose identity cannot be sufficiently established.

Possible treatment may include:

Resolve

Restrict

Exclude

or:

Escalate for further assessment

depending upon the simulation context.

No Anonymous Material Input Principle

SIMULOS® establishes:

A materially influential input should not enter governed simulation execution anonymously.

Input Equivalence

Two inputs should not be considered equivalent merely because: their values match, their names match, their formats match, or they occupy the same software field.

Input equivalence may depend upon: semantics, origin, unit, population, environment, time, transformation, and simulation role.

Input Substitution

An identified input may later be substituted.

For example:

Live Sensor Input

may be replaced with:

Historical Replay Input

The replacement should receive its own identity and classification.

The simulation record should preserve:

Original Intended Input → Substitution → Actual Consumed Input Actual Consumed Input

SIIM distinguishes:

Planned Input

from:

Actual Consumed Input.

This distinction becomes critical when execution does not consume exactly what was originally configured.

Input Set Membership

Each material Simulation Input may belong to one or more governed Simulation Input Sets.

Conceptually:

Simulation Input Set SIS-001

may contain:

SI-001 + SI-002 + SI-003 + SI-004

Input-set governance enables reconstruction of the complete information basis of an execution.

Input Set Identifier

A Simulation Input Set may receive a persistent identifier.

This allows:

Execution → Input Set → Individual Inputs

to remain traceable.

Input Set Composition

The composition of an Input Set should identify materially relevant: inputs, input versions, classifications, states, source references, and applicable simulation configuration.

Input Version Distinction

Where materially changed information exists under the same logical input identity, versions should remain distinguishable.

For example:

Weather Dataset v3

should not silently overwrite:

Weather Dataset v2

where previous executions relied upon v2.

Input Change

A material input may change because: source data changes, values change, origin changes, classification changes, format changes, unit changes, derivation changes, or its simulation role changes.

The changed input should remain distinguishable from the previous state where required for reconstruction.

Input Identity Continuity

Input Identity Continuity preserves the relationship between successive states or versions of the same logical input.

Conceptually:

SI-001 v1 → SI-001 v2 → SI-001 v3

while preventing materially different objects from being falsely treated as one unchanged input.

Point-in-Time Input Identity

SIIM supports reconstruction of:

Which exact input existed at the time of a particular simulation execution?

This is essential where input data changes after execution.

No Retrospective Input Replacement Principle

SIMULOS® establishes:

Updating an input must not silently rewrite the historical input identity of completed simulation executions.

Input Traceability

SIIM establishes the traceability foundation:

Input Requirement

↓

Simulation Input

↓

Origin / Source Reference

↓

Classification

↓

Simulation Role

↓

Model / Environment / Scenario Binding

↓

Input Set

↓

Simulation Execution Reference

This chain enables later governance of simulation behavior and outputs.

Input Identification Status

A material Simulation Input may receive an identification status such as:

Identified

Conditionally Identified

Ambiguous

Unidentified

Superseded

or:

Excluded

The status concerns input identity.

It does not establish representativeness, temporal suitability, uncertainty, or general validity.

Identification Sufficiency

Input identification is sufficient when the information necessary to distinguish and trace the material input is adequate for its intended simulation role.

The required identification depth may vary with: simulation consequence, input influence, system complexity, simulation purpose, and downstream reliance.

Identification Failure

An Input Identification Failure occurs where a material input cannot be sufficiently distinguished or traced for governed use.

Possible causes include: missing origin, ambiguous semantics, unknown units, indistinguishable versions, undocumented substitution, unknown derivation, or inability to determine the actual consumed input.

Identification Failure ≠ Data Invalidity

Failure to identify an input sufficiently does not prove the underlying information is false.

It establishes that the information cannot currently support the required level of governed simulation traceability.

Simulation Input Register

SIIM may maintain a Simulation Input Register containing material Simulation Inputs and their governance attributes.

A register entry may contain: Simulation Input Identifier, input name, description, classification, origin, source reference, unit, format, semantic definition, input state, simulation role, applicable object, model binding, environment binding, scenario reference, input-set membership, version, and identification status.

Minimum Implementation Framework

1. Identify Input Requirements

Determine which information the governed simulation requires. 2. Discover Material Proposed Inputs

Identify all materially influential information proposed for simulation use.

3. Assign Simulation Input Identity

Create a persistent identifier for each material input where appropriate. 4.

Define Input Meaning

Record sufficient: name, semantics, unit, format, and represented quantity or state.

5. Classify Input Origin and Type

Determine whether the input is: observed, measured, historical, live, estimated, derived, synthetic, human-supplied, externally supplied, model-generated, scenario-specified, or another defined class.

6. Preserve Origin and Source Reference

Bind the input to its identifiable origin and source where available. 7. Bind the Input to Simulation Role

Establish which: Simulation Object, model component, environment element, configuration, or scenario context

the input may influence. 8. Identify Dependencies and Composite Inputs

Preserve material relationships between inputs. 9. Detect Ambiguity, Duplication, and Substitution

Identify materially unclear, duplicated, replaced, or unexpectedly consumed inputs. 10. Establish Input Set Membership

Bind the identified inputs to the applicable Simulation Input Set. 11. Assign Identification Status

Determine whether each material input is sufficiently identified for further SIDS governance. 12. Preserve Input Traceability

Maintain the chain from input requirement and origin through simulation role and eventual execution use.

Governance Outputs

SIIM may produce: Simulation Input Requirement Record, Simulation Input Identifier, Simulation Input Record, Simulation Input Classification, Simulation Input Origin Record, Simulation Input Source Reference, Simulation Input Semantic Definition, Simulation Input Unit Record, Simulation Input Format Record, Simulation Input State Record, Simulation Input Role Record, Input-to-Simulation Object Binding, Input-to-Model Binding, Input-to-Environment Binding, Input-to-Configuration Binding, Input-to-Scenario Reference, Simulation Input Dependency Map, Composite Input Record, Duplicate Input Record, Ambiguous Input Record, Unidentified Input Record, Synthetic Input Identification Record, Derived Input Identification Record, Estimated Input Identification Record, Input Substitution Record, Actual Consumed Input Record, Simulation Input Set, Simulation Input Set Identifier, Input Set Composition Record, Input Version Record, Input Identity Continuity Record, Point-in-Time Input Identity Record, Input Identification Status, Input Identification Failure Record, Simulation Input Register, and

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous vehicle simulation consumes: camera data, radar measurements, weather information, road geometry, vehicle state, synthetic pedestrian behavior, and traffic signals.

Application

SIIM identifies each material input and distinguishes:

Measured Radar Input

from:

Synthetic Pedestrian Input

and:

Historical Weather Dataset

from:

Current Vehicle-State Input.

Each input is bound to the relevant model or environment component. Result

The simulation possesses an identifiable input basis before representativeness, transformation, temporal applicability, and uncertainty are assessed. Use Case 2 — Spacecraft Landing Simulation

Scenario

A spacecraft landing simulation uses: terrain elevation data, gravitational model outputs, estimated atmospheric density, vehicle telemetry, engine-performance measurements, and synthetic failure conditions.

Application

SIIM identifies the origin, classification, meaning, units, and simulation role of each material input.

The module prevents:

Estimated Atmospheric Density

from becoming indistinguishable from:

Direct Atmospheric Measurement

and preserves synthetic failure conditions as synthetic inputs. Result

Every materially influential input can be traced to its origin and simulation role, providing a controlled basis for subsequent SIMULOS® input governance.

Architectural Position

Simulation Input Identification is the first internal governance space of the Simulation Input & Data Standard (SIDS).

The complete SIDS progression is:

Simulation Input Identification

↓

Simulation Data Representativeness

↓

Simulation Data Transformation

↓

Simulation Temporal Data

↓

Simulation Input Uncertainty

SIIM therefore establishes the identity layer upon which all subsequent SIDS governance depends.

Representativeness cannot be meaningfully assessed until the input being assessed is known.

Transformation cannot be governed unless the source and resulting input can be distinguished.

Temporal applicability cannot be evaluated without an identifiable information object.

Input uncertainty cannot remain attached to an input whose identity is unstable.

Foundational Principle

A simulation cannot govern the suitability of information it cannot first identify, distinguish, classify, and trace to the role through which that information can influence simulated behavior.

Canonical Closing Statement

Simulation Input Identification Module establishes the identity and traceability foundation of the Simulation Input & Data Standard by governing how materially influential data, values, signals, measurements, states, distributions, events, synthetic information, derived information, and other simulation inputs are discovered, identified, classified, source-referenced, distinguished, versioned, bound to their simulation roles, assembled into controlled Input Sets, and preserved across execution history. Through explicit separation of observed, measured, estimated, derived, synthetic, historical, live, human-supplied, externally supplied, model-generated, and scenario-specified information, SIIM prevents anonymous, ambiguous, substituted, or semantically unclear inputs from silently entering governed simulation activity and establishes the traceability basis required for representativeness, transformation, temporal, uncertainty, execution, and downstream simulation governance.

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